

Civil Engineering

Calculus and Differential Methods

Revision Questions

Use these questions for CAT and summative assessment preparation.

Formula Practice

1. Apply the formula below in a technical example.

$$\frac{d}{dx}(ax^n) = anx^{n-1}$$

2. Apply the formula below in a technical example.

$$\int ax^n dx = \frac{ax^{n+1}}{n+1} + C$$

3. Apply the formula below in a technical example.

$$\frac{dy}{dx} + P(x)y = Q(x)$$

4. Apply the formula below in a technical example.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Applied Practice

5. Find the gradient function of a road profile.
6. Determine area under a drainage channel curve.
7. Apply Newton-Raphson method to find a root.